

WEIL POSITIVITY ON THE WINDOW OF WIDTH $\log 12$, AND A LOWER BOUND FOR THE PROLATE EIGENVALUE DEFECT

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ABSTRACT. Weil’s criterion states that the Riemann Hypothesis holds if and only if Weil’s quadratic form Q is nonnegative on all admissible test functions. Restricting the support to a fixed interval $[-L, L]$ gives a weaker, unconditional question. We prove that $Q(f) \geq 2^{-1518} \|f\|_2^2$ for every complex C^2 function f supported in $[-\frac{1}{2} \log 12, \frac{1}{2} \log 12]$; the half-width $\frac{1}{2} \log 12 \approx 1.2425$ exceeds that of the previous proofs we know of, the largest being $\frac{1}{2} \log 10$ (our previous paper). The new analytic ingredient is an explicit lower bound for the defect of the top eigenvalue of time and frequency limiting: $1 - \lambda_0(c) \geq e^{-2(c+1)}/(4(c+1)(c+4))$ for all $c > 0$, which has the correct exponential rate e^{-2c} . It replaces the out-of-band estimate of our previous paper, which was proved under the convenient hypothesis $L < 7/6$, and it improves the constant on the window of half-width $\frac{1}{2} \log 10$ from 2^{-49162} to 2^{-876} without recomputing the finite matrices. The proof follows Liu’s bounded-comparison framework and reduces positivity to a finite matrix inequality of size 416 in each parity, which we certify in Arb ball arithmetic with all quadrature, rounding and inversion errors paid: the finite margin is about 1.21×10^{-54} (even) and 1.59×10^{-50} (odd), while the total error is below 10^{-75} . It is an external certificate, not a formal proof. The result concerns one fixed window and does not address the Riemann Hypothesis.

Version 1. Zenodo, doi:10.5281/zenodo.23261744. Certificate data and code: release v1.5.0 of <https://github.com/moririyota/prolate-weil-bounds>, doi:10.5281/zenodo.23092422. Licence: CC BY 4.0.

Status. Analytic proof plus an external interval-arithmetic certificate (Arb/FLINT). The lower bound for the prolate defect, the general tail estimate and the certificate passed two independent AI reviews each, including an independent reimplementing of the final check; no blocking issues remained. An independent review of this manuscript led to local corrections of proof text and citations. Not formalized in Lean. Not yet reviewed by independent human experts.

1. INTRODUCTION

1.1. Background. Let Q be Weil’s quadratic form [19, 4]. Weil’s criterion states that RH is equivalent to $Q(f) \geq 0$ for all admissible f . If the support of f is restricted to $[-L, L]$, only prime powers $n < e^{2L}$ contribute and the question can be decided unconditionally for small L . Positivity is known for $L = \frac{1}{2} \log 2$ (Yoshida [20]), $L = 0.8$ (Zhu [21]), $L = \frac{1}{2} \log 5$ (Lean 4, [11]), $L = 17/16$ (Liu [10], unrefered) and $L = \frac{1}{2} \log 10$ ([14]). For $\mu = e^{2L}$, the bottom $\lambda_{\min}$ of the spectrum of the restricted form satisfies $\lambda_{\min} \leq 3.50 \times 10^{24} \mu^{9/2} \log \mu e^{-4\pi\mu}$ for $50 \leq \mu \leq 10^4$, and $\lambda_{\min} \leq 7.58 \times 10^{24} \mu^5 (\log \mu)^3 e^{-4\pi\mu}$ for every $\mu \geq 50$ [13, Theorem K]. These are upper bounds only, but they show that each enlargement of the window must resolve a much smaller margin.

In [14] the high frequencies were controlled by a lower bound on the out-of-band mass, $I - B_{\text{band}} \succeq \delta I$, proved by Lagrange interpolation. That argument was stated under the convenient sufficient condition $L < 7/6$ (that is, $e^{2L} < 10.3$) and gave only $\delta = 2^{-49158}$, while the best constant is the defect $1 - \lambda_0(\Omega L)$ of time and frequency limiting, of size $e^{-2\Omega L}$ up to powers.

1.2. Results. For $L > 0$, $f \in C^2(\mathbb{R}; \mathbb{C})$ with $\text{supp } f \subset [-L, L]$, put $F_f(z) = \int f(x)e^{izx} dx$, $a(t) = \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi$ and

$$(1.1) \quad \begin{aligned} Q(f) = & 2 \left| \int f(x) \cosh \frac{x}{2} dx \right|^2 - 2 \left| \int f(x) \sinh \frac{x}{2} dx \right|^2 + \frac{1}{2\pi} \int_{\mathbb{R}} a(t) |F_f(t)|^2 dt \\ & - 2 \sum_{n < e^{2L}} \frac{\Lambda(n)}{\sqrt{n}} \text{Re} \int_{\mathbb{R}} f(x) \overline{f(x - \log n)} dx. \end{aligned}$$

By the Guinand–Weil explicit formula, $Q(f) = \text{Re} \sum_{\rho} m_{\rho} F_f(\gamma_{\rho}) \overline{F_f(\overline{\gamma_{\rho}})}$ with $\gamma_{\rho} = (\rho - \frac{1}{2})/i$; RH is not assumed. The derivation, the convergence of the zero sum, and the reduction to real symmetric operators on each parity are as in [14, Section 2]; nothing there depends on L . For $L = \frac{1}{2} \log 12$ the prime powers are $n \in \{2, 3, 4, 5, 7, 8, 9, 11\}$.

Theorem M. *For every $f \in C^2(\mathbb{R}; \mathbb{C})$ with $\text{supp } f \subset [-\frac{1}{2} \log 12, \frac{1}{2} \log 12]$,*

$$Q(f) \geq 2^{-1518} \int_{\mathbb{R}} |f(x)|^2 dx.$$

Theorem N. *Let $\lambda_0(c)$ be the largest eigenvalue of the operator on $L^2(-1, 1)$ with kernel $\sin(c(x - y))/(\pi(x - y))$. For every $c > 0$,*

$$1 - \lambda_0(c) \geq \frac{e^{-2(c+1)}}{4(c+1)(c+4)}.$$

Moreover the normalized positive ground state satisfies $\psi_{0,c}(1)^2 \geq e^{-2c}/(4c+12)$.

Corollary 1.1. *The constant in Theorem L of [14] can be replaced by 2^{-876} : $Q(f) \geq 2^{-876} \|f\|^2$ for every $f \in C^2(\mathbb{R}; \mathbb{C})$ supported in $[-\frac{1}{2} \log 10, \frac{1}{2} \log 10]$.*

Theorem M is proved by an ordinary analytic argument together with an external certificate in ball arithmetic (Arb/FLINT [1]), produced and checked by separate programs. It is not a formal proof.

reference	half-width L	e^{2L}	method
Yoshida [20]	$\frac{1}{2} \log 2 \approx 0.347$	2	analytic + computer-assisted
Zhu [21]	0.8	4.95	interval certificate
[11]	$\frac{1}{2} \log 5 \approx 0.805$	5	Lean 4 (kernel-checked)
Liu [10]	$17/16 = 1.0625$	8.37	interval certificate (unrefereed)
[14]	$\frac{1}{2} \log 10 \approx 1.151$	10	interval certificate
Theorem M	$\frac{1}{2} \log 12 \approx 1.242$	12	interval certificate

TABLE 1. Unconditional positivity on windows $[-L, L]$. The approach of Connes and Consani [5] works under vanishing conditions on the Fourier transform and is not listed.

1.3. Method and what is new.

- Theorem N: a comparison of the ground state with $x^{-1/2} I_0(c\sqrt{1-x^2})$, combined with Slepian's identity. The exponential rate e^{-2c} is sharp (Fuchs [7]); the polynomial factor is c^{-2} instead of $\sqrt{c}$. A lower bound $C^{-1} e^{-2Cc/\pi}$ with an unspecified constant follows from the uncertainty principle of Nazarov and Jaming [15, 8]; explicit endpoint bounds of Bonami and Karoui [2, 3] require conditions that the ground state does not satisfy. We do not claim that Theorem N is new.
- A tail inequality for general L and Ω , with the projection errors written as functions of N and of the ellipse parameter ρ (Section 3).

- The window $\frac{1}{2} \log 12$, where the prime 11 enters, with $\|C_p\| \leq \frac{57}{16}$ and $\Omega = 416$; the finite condition is certified at $N = 832$; at $N = 640$ the computed comparison matrix was numerically negative in both parities (Section 5).

2. THE DEFECT OF THE TOP PROLATE EIGENVALUE

Let Q_c be the operator on $L^2(-1, 1)$ with kernel $\sin(c(x - y))/(\pi(x - y))$ and $D_c = -\frac{d}{dx}(1 - x^2)\frac{d}{dx} + c^2x^2$. We use the following classical facts (see also [18, 17]). The operators Q_c and D_c have a common complete system of real eigenfunctions $\psi_{n,c}$, and the largest eigenvalue $\lambda_0(c)$ of Q_c and the smallest eigenvalue $\chi_0(c)$ of D_c belong to the same simple eigenfunction $\psi = \psi_{0,c}$ [9, Section 2.1]. This eigenfunction is even and has no zeros in $(-1, 1)$ [16, Theorems 1 and 3]. The finite Fourier eigenfunction relation gives an entire extension, and the endpoint values are nonzero by uniqueness of the analytic solution at the regular singular endpoints. We normalize $\psi > 0$ and $\int_{-1}^1 \psi^2 = 1$. Slepian's identity, with this normalization, is proved in [12, Lemma 2.1]:

$$(2.1) \quad \lambda'_0(c) = \frac{2\lambda_0(c)\psi_{0,c}(1)^2}{c}.$$

In particular λ_0 is increasing. Write $p(x) = 1 - x^2$ and $A = \psi(1) > 0$. The eigenvalue equation is $(p\psi')' = (c^2x^2 - \chi_0)\psi$, and the constant test function gives $0 < \chi_0 \leq c^2/3$.

Lemma 2.1 (centre). $\psi' < 0$ on $(0, 1)$, and $\psi(a) \geq \frac{5}{6}\psi(0)$ for $a = 1/(2(c + 1))$.

Proof. $p\psi'$ vanishes at 0 (parity) and at 1 (regularity), and $(p\psi')'$ changes sign once on $(0, 1)$, at $x = \sqrt{\chi_0}/c$; hence $p\psi' < 0$ on $(0, 1)$. Since $\psi \leq \psi(0)$ there, $-p(x)\psi'(x) = \int_0^x (\chi_0 - c^2t^2)\psi dt \leq \chi_0\psi(0)x \leq c^2\psi(0)x$. Integrating, $\psi(a) \geq \psi(0)(1 + \frac{c^2}{2} \log(1 - a^2))$, and $-\frac{c^2}{2} \log(1 - a^2) \leq \frac{c^2a^2}{2(1-a^2)} \leq \frac{1/4}{3/2} = \frac{1}{6}$. $\square$

Lemma 2.2 (comparison). Let $Y(x) = x^{-1/2}I_0(c\sqrt{1-x^2})$ for $0 < x \leq 1$, where I_0 is the modified Bessel function. Then

$$(pY')' = \left(c^2x^2 + \frac{3}{4x^2} + \frac{1}{4}\right)Y,$$

and $\psi(x) \leq AY(x)$ for $0 < x \leq 1$.

Proof. Put $w = I_0(c\sqrt{p})$. From the modified Bessel equation [6, 10.25.1], $(pw')' = c^2x^2w - c\sqrt{p}I_1(c\sqrt{p})$ and $w' = -cxI_1(c\sqrt{p})/\sqrt{p}$. With $Y = x^{-1/2}w$, the cross term $2p(x^{-1/2})'w' = c\sqrt{p}I_1(c\sqrt{p})x^{-1/2}$ cancels the last term, and $(p(x^{-1/2})')' = (\frac{3}{4x^2} + \frac{1}{4})x^{-1/2}$ gives the identity. Moreover $Y(1) = 1$, and $pY' \rightarrow 0$ as $x \rightarrow 1$.

Let $d = Y - \psi/A$. Then $d(1) = 0$. Evaluating the two differential equations at $x = 1$ gives $Y'(1) = -(c^2 + 1)/2$ and $\psi'(1) = -(c^2 - \chi_0)A/2$, so $d'(1) = -(1 + \chi_0)/2 < 0$ and $d > 0$ near 1. Also

$$(pd')' = c^2x^2d + \left(\frac{3}{4x^2} + \frac{1}{4}\right)Y + \frac{\chi_0}{A}\psi.$$

Suppose d vanishes in $(0, 1)$ and let x_1 be its largest zero there. On $(x_1, 1)$ the right side is positive, so $pd'(x) = -\int_x^1 (pd')' < 0$, and $d(x_1) = -\int_{x_1}^1 d' > 0$, a contradiction. $\square$

Proof of Theorem N. By [6, 10.32.1], $I_0(z) \leq e^z$ for $z \geq 0$, so Lemma 2.2 gives $\psi(x)^2 \leq A^2x^{-1}e^{2c}$ on $(0, 1]$. On $[0, a]$, Lemma 2.1 gives $\psi^2 \leq \psi(0)^2 \leq 4\psi(a)^2 \leq 4A^2a^{-1}e^{2c}$. Hence

$$1 = 2 \int_0^1 \psi^2 \leq 2A^2e^{2c}(4 + \log \frac{1}{a}) = A^2e^{2c}(8 + 2\log(2(c + 1))) \leq A^2e^{2c}(4c + 12),$$

which is the bound for $\psi_{0,c}(1)^2$. If $\lambda_0(c) \leq \frac{1}{2}$, then $1 - \lambda_0(c) \geq \frac{1}{2}$, which exceeds the claimed bound. Otherwise $\lambda_0(s) > \frac{1}{2}$ for $s \geq c$, and by (2.1)

$$1 - \lambda_0(c) \geq \lambda_0(c+1) - \lambda_0(c) = \int_c^{c+1} \frac{2\lambda_0(s)\psi_{0,s}(1)^2}{s} ds \geq \int_c^{c+1} \frac{e^{-2s}}{s(4s+12)} ds \geq \frac{e^{-2(c+1)}}{4(c+1)(c+4)}. \quad \square$$

Remark 2.3. (i) For a window $[-L, L]$ and the band $|t| \leq \Omega$ (with $F_f(t) = \int f(x)e^{itx}dx$), scaling $x = Ls$ shows that $\frac{1}{2\pi} \int_{|t| > \Omega} |F_f|^2 \geq (1 - \lambda_0(\Omega L))\|f\|^2$, so Theorem N gives the out-of-band bound $I - B_{\text{band}} \succeq d(\Omega L)I$ with $d(c) = e^{-2(c+1)}/(4(c+1)(c+4))$, for every L . (ii) By Fuchs [7], $1 - \lambda_0(c) \sim 4\sqrt{\pi}c^{1/2}e^{-2c}$, so Theorem N has the correct exponential rate and loses a factor of order $c^{5/2}$. A Legendre–Galerkin computation for $0.5 \leq c \leq 20$ gives ratios between 3.7×10^2 and 4.6×10^5 (numerical, not used).

3. THE BOUNDED COMPARISON FOR GENERAL WINDOWS

Throughout this section $L > 0$ is fixed, $I = [-L, L]$, $H = L^2(I; \mathbb{C})$, and $\Omega > 0$, $m > 0$ are parameters. The argument is that of [14, Sections 3–4] with every constant written in terms of L , Ω , m and the truncation order N ; it follows the framework of Liu [10].

3.1. The prime operator. For $a \in \mathbb{R}$ let $S_a f(x) = \mathbf{1}_I(x-a)f(x-a)$, $C_p = \sum_{n < e^{2L}} \Lambda(n)n^{-1/2}(S_{\log n} + S_{-\log n})$, so that the prime term of (1.1) is $-\langle f, C_p f \rangle$. For a positive even weight w the weighted Schur inequality gives $\|C_p\| \leq \text{ess sup } R_w$, where

$$R_w(x) = \sum_{n < e^{2L}} \frac{\Lambda(n)}{\sqrt{n}} \frac{\mathbf{1}_I(x + \log n)w(x + \log n) + \mathbf{1}_I(x - \log n)w(x - \log n)}{w(x)}.$$

In the variable $q = e^{2x}$, $x \geq 0$, the shift by $+\log n$ is active for $q < e^{2L}/n^2$ and the shift by $-\log n$ for $q > n^2e^{-2L}$. If $\|C_p\| \leq s$, put $\gamma = s + m$ and $M = \gamma I - C_p$; then $mI \preceq M \preceq bI$ with $b = 2s + m$.

3.2. The tail. Let B_{band} have kernel $\sin(\Omega(x-y))/(\pi(x-y))$ on I , and

$$K = 2|\cosh \frac{x}{2}\rangle\langle\cosh \frac{x}{2}| - 2|\sinh \frac{x}{2}\rangle\langle\sinh \frac{x}{2}| + \frac{1}{2\pi} \int_{-\Omega}^{\Omega} (a(t) - \gamma)|e_t\rangle\langle e_t| dt, \quad e_t(x) = e^{itx}.$$

Then $Q(f) = \langle f, (M + K)f \rangle + T_{\text{tail}}(f)$ with $T_{\text{tail}}(f) = \frac{1}{2\pi} \int_{|t| > \Omega} (a(t) - \gamma)|F_f(t)|^2 dt$. By the proof of Lemma 3.3 of [14], $a(t) \geq \log \frac{t}{2\pi} - \frac{1}{t}$ for $t \geq \frac{15}{4}$; the right side increases with t , so

$$(3.1) \quad a(t) - \gamma \geq C := \log \frac{\Omega}{2\pi} - \frac{1}{\Omega} - \gamma \quad (|t| \geq \Omega \geq \frac{15}{4}).$$

Proposition 3.1 (tail inequality). *Assume $C > 0$, let $c = \Omega L > 2$, and let $0 < \delta \leq d(c) = e^{-2(c+1)}/(4(c+1)(c+4))$. Put $S = I - B_{\text{band}} - \delta I$, $v_0 = (2L)^{-1/2}$, $v_1 = \sqrt{3/(2L)}x/L$, $h_j = Sv_j$,*

$$E_0 = \frac{c+1}{\pi c^2}, \quad E_1 = \frac{3c^2 + 6c + 2}{\pi c^3},$$

and let $u_0, u_1 \geq 0$ satisfy $u_j E_j \leq C$. Then, with $U = u_0|h_0\rangle\langle h_0| + u_1|h_1\rangle\langle h_1|$,

$$T_{\text{tail}}(f) \geq C\delta\|f\|^2 + \langle f, Uf \rangle.$$

Proof. By Theorem N and Remark 2.3(i), $I - B_{\text{band}} \succeq d(c)I \succeq \delta I$, so $0 \preceq S \preceq I$, and (3.1) gives $T_{\text{tail}}(f) \geq C\delta\|f\|^2 + C\langle f, Sf \rangle$. As in [14, Proposition 3.4], $e_j = \langle v_j, (I - B_{\text{band}})v_j \rangle$ satisfies $\frac{c-1}{\pi c^2} \leq e_0 \leq E_0$ and $\frac{3(c-2)}{\pi c^2} \leq e_1 \leq E_1$; these bounds come from integration by parts and hold for every $c > 2$. In fact $1 - \lambda_0(c) > d(c)$: this is immediate if $\lambda_0(c) \leq \frac{1}{2}$; otherwise the first inequality in the proof of Theorem N is strict, since $\lambda_0(c+1) < 1$. Thus $S \succeq (1 - \lambda_0(c) - \delta)I \succ 0$, and $\langle v_j, Sv_j \rangle = e_j - \delta > 0$ for both $j = 0, 1$. The vectors v_0, v_1 have opposite parities and S

commutes with the reflection, so the Cauchy–Schwarz inequality for the form $\langle \cdot, S \cdot \rangle$ on each parity gives $\langle f, Sf \rangle \geq \sum_j |\langle h_j, f \rangle|^2 / (e_j - \delta)$, and $C/(e_j - \delta) \geq C/E_j \geq u_j$. $\square$

With $V = K + U$ we obtain, for admissible f ,

$$(3.2) \quad Q(f) \geq C\delta \|f\|^2 + \langle f, (M + V)f \rangle,$$

and it suffices to prove $M + V \succeq 0$ on H .

3.3. Reduction to a finite matrix. Let $N \geq 2$ be an integer, counting both parities, and let $v_j(x) = \sqrt{(2j+1)/(2L)} P_j(x/L)$, $E : \mathbb{C}^N \rightarrow H$, $Ez = \sum_{j < N} z_j v_j$, $P = EE^*$, and

$$A = E^*ME, \quad B = E^*M^2E, \quad J = E^*VE, \quad G = E^*M^{-1}E, \quad D_M = B - A^2 \succeq 0.$$

Let $R = V - EJE^*$, and suppose $\|(I - P)RP\| \leq e$, $(I - P)R(I - P) \succeq -n$ and $\|(I - P)R(I - P)\| \leq h$ with $n < m$. With

$$\alpha = e + \frac{2e^2}{m - n}, \quad \beta = \frac{e + n}{m^2} + \frac{2h^2}{m^2(m - n)},$$

the block elimination of [14, (4.1)] (with $\theta = \chi = 1$; the proof uses only $mI \preceq M$ and $n < m$) gives

$$(3.3) \quad G^{-1} + J \succeq \alpha I + \beta D_M \implies M + V \succeq 0 \text{ on } H.$$

For every matrix X , $\mathcal{W}(X) = X + X^* - X^*AX + m^{-1}(I - AX - X^*A + X^*BX)$ satisfies $\mathcal{W}(X) \succeq G$ [14, Section 4]. So it suffices to certify, in each parity,

$$(3.4) \quad T = \mathcal{W}(X)^{-1} + J - \alpha I - \beta(B - A^2) \succ 0.$$

Lemma 3.2 (projection errors). *Let $\rho > 1$, $c = \Omega L$, and*

$$r_N^2 = \frac{8L\rho^2}{(\rho - 1)^2} e^{c(\rho - \rho^{-1})} \rho^{-2N}, \quad K_0 = \frac{\Omega}{\pi} \max(|a(0) - \gamma|, |a(\Omega) - \gamma|),$$

$H_p = \sqrt{2L} \cosh \frac{L}{2}$ and $q_N = \sqrt{2L} e^{L/2} (L/2)^N / N!$. Then one may take

$$e = \left[K_0 \sqrt{2L} + \sqrt{\Omega/\pi} (u_0 \sqrt{E_0} + u_1 \sqrt{E_1}) \right] r_N + 4H_p q_N, \quad n = K_0 r_N^2 + 4q_N^2, \quad h = \left[K_0 + \frac{\Omega}{\pi} (u_0 + u_1) \right] r_N^2 + 4q_N^2.$$

Proof. For $|t| \leq \Omega$, e^{itx} on I is, after scaling to $[-1, 1]$, bounded by $e^{c(\rho - \rho^{-1})/2}$ on the Bernstein ellipse E_ρ . Hence its Chebyshev coefficients are at most $2H\rho^{-k}$, where $H = e^{c(\rho - \rho^{-1})/2}$, and the truncation of degree $N - 1$ has sup-norm error at most $2H\rho^{-N}/(1 - \rho^{-1}) = 2H\rho^{1-N}/(\rho - 1)$. Since the L^2 projection is optimal and $\|1\|_{L^2(I)}^2 = 2L$, this gives $\|(I - P)e_t\| \leq r_N$. The function a is increasing on $t \geq 0$: in the series for ψ , each term $\text{Re}(\frac{1}{4} + \frac{it}{2} + k)^{-1}$ decreases in t . Therefore $\frac{1}{2\pi} \int_{-\Omega}^{\Omega} |a - \gamma| \leq K_0$. The band part of R is then bounded by $K_0 \sqrt{2L} r_N$ off the diagonal block, and by $K_0 r_N^2$ in the complementary block (from below and in norm).

For U : $(I - P)h_j = -(I - P)B_{\text{band}}v_j = -\frac{1}{2\pi} \int_{-\Omega}^{\Omega} \hat{v}_j(t)(I - P)e_t dt$, with norm at most $r_N \sqrt{\Omega/\pi}$ by Cauchy–Schwarz and Plancherel. Also $\|h_j\|^2 \leq \langle v_j, S v_j \rangle \leq E_j$, and $U \succeq 0$, so U contributes nothing to n .

For the poles: the Taylor remainder of $e^{\pm x/2}$ of order N gives $\|(I - P) \cosh \frac{x}{2}\|, \|(I - P) \sinh \frac{x}{2}\| \leq q_N$, and both norms are at most H_p . $\square$

4. THE WINDOW $\frac{1}{2} \log 12$

Let $L = \frac{1}{2} \log 12$, so $e^{2L} = 12$ and the prime powers are 2, 3, 4, 5, 7, 8, 9, 11.

Prime operator. With $w(x) = 1 + \frac{19}{40}(x/L)^2$, the breakpoints on $x \geq 0$ are $q \in \{1, \frac{4}{3}, \frac{25}{12}, 3, \frac{49}{12}, \frac{16}{3}, \frac{27}{4}, \frac{121}{12}, 12\}$. Bounding R_w in ball arithmetic on $8 \cdot 256 = 2048$ closed subintervals gives $R_w < 3.5557$, hence $\|C_p\| \leq s = \frac{57}{16}$. We take $m = \frac{1}{2}$, so $\gamma = \frac{65}{16}$ and $b = \frac{61}{8}$.

Tail. With $\Omega = 416$ we get $c = \Omega L \approx 516.86$ and $C \approx 0.12790 > \frac{1}{8}$. Moreover $-\log_2 d(c) = 1514.27\dots$, so we take $\delta = 2^{-1515}$ and $\tau = \delta/8 = 2^{-1518} < C\delta$. The coefficients $u_0 = 207$, $u_1 = 68$ satisfy $u_j E_j < C$.

Projection errors. With $N = 832$ (that is, 416 modes in each parity) and $\rho = 3$, Lemma 3.2 gives, in ball arithmetic,

$$e < 2.08 \times 10^{-94} < 2^{-311}, \quad n < 1.27 \times 10^{-191} < 2^{-634}, \quad h < 3.82 \times 10^{-190} < 2^{-629}.$$

Then $n < m$, and $\alpha + \beta b^2 < 2^{-303}$ (checked with exact rationals).

Remark 4.1. At $N = 640$ the midpoint value of the smallest eigenvalue of $\mathcal{W}(X_*)^{-1} + J$ was negative in both parities (-1.99×10^{-46} even, -4.89×10^{-50} odd). The negative directions are concentrated on Legendre degrees below 32, and the simple compression $A + J$ is positive in the same directions. So the failure comes from the loss in the inverse bound, not from the Weil form. At $N = 832$ the condition holds. For comparison, our Legendre–Galerkin computation of the bottom of the spectrum of Q on this window gives about 4.05×10^{-54} (numerical, not used in the proof; cf. [13]).

5. SOURCE MATRICES, QUADRATURE AND THE CERTIFICATE

A and B. After splitting I at the support boundaries of the shifts, the entries are integrals of polynomials of degree at most 1662; Gauss–Legendre quadrature with 834 nodes is exact, so only rounding errors occur. These are enclosed by the balls, and $\epsilon_A = \epsilon_B = 2^{-400}$ covers them together with the distance to the rational centres.

J. In each parity the band and gamma-weighted integrals are computed on the panels $[0, \frac{1}{2}]$, $[\frac{1}{2}, 1]$, $[1, 2]$, $\dots$, $[128, 256]$, $[256, 416]$ with 256 Gauss nodes each, from the representation $\int_I v_j e^{itx} dx = \sqrt{2L(2j+1)} i^j j_j(Lt)$.

Lemma 5.1 (quadrature). *On the Bernstein ellipses $\rho = 2$ of the panels, $|\operatorname{Im} z| \leq 60$ and $|z| \leq 436$. There $|b_i b_j| \leq 2Le^{120L} < \frac{5}{2}e^{150}$, and $g(z) = \frac{1}{2}[\psi(\frac{1}{4} + \frac{iz}{2}) + \psi(\frac{1}{4} - \frac{iz}{2})] - \log \pi - \frac{65}{16}$ is holomorphic with $|g| < 1100$. Consequently the quadrature error of J , including the pole and U terms, is below 2^{-263} in operator norm.*

Proof. The longest panel $[256, 416]$ has half-length 80, so its ellipse has semi-minor axis $80 \cdot \frac{3}{4} = 60$, and every point satisfies $|z| \leq 336 + 100$. For $w = \frac{1}{4} \pm \frac{iz}{2}$: on the three panels of length at most 1, $\operatorname{Re} w \geq \frac{1}{16}$; on the others, $\operatorname{Re} z \geq \frac{7}{4}$, so $|\operatorname{Im} w| \geq \frac{7}{8}$. Hence $|w + k| \geq \frac{1}{16}$ for all integers $k \geq 0$.

After 31 applications of the recurrence, $v = w + 31$ has $\operatorname{Re} v \geq \frac{5}{4}$ and $|v - 1| < 249$, and the series $\psi(v) = -\gamma_E + (v - 1) \sum_{k \geq 0} \frac{1}{(k+1)(k+v)}$ [6, 5.7.6] gives $|\psi(v)| < 499$. The recurrence costs at most $31 \cdot 16$.

A Chebyshev truncation of degree 511 and the exactness of 256-point Gauss up to degree 511 give the panel error $4\ell H 2^{-511}$. The sum of the panel lengths is 416, and the operator norm is at most 416 times the entrywise bound. The pole coefficients use 872 nodes and the Taylor polynomials of degree 912, with remainder $2(5/8)^{913}/913!$. For U , $\|E^* S v_j\| \leq 1$. Adding the three contributions in ball arithmetic gives $5.45 \times 10^{-80} < 2^{-263}$. $\square$

The constant δ . The saved matrix J was computed with $\delta_{\text{old}} = d(c)$ (enclosed in a ball), while the theorem uses $\delta = 2^{-1515} \leq d(c)$. With $t = \delta_{\text{old}} - \delta \in [0, 2^{-1514})$ we have $h_j^{\text{new}} = h_j^{\text{old}} + t v_j$, so

$\|U^{\text{new}} - U^{\text{old}}\| \leq (u_0 + u_1)(2t + t^2) < 2^{-1504}$. This is added to the error of J . The pieces $(I - P)h_j$ do not change, so Lemma 3.2 still applies.

Certificate. As in [14, Section 6], for each parity the proposer outputs rational matrices with denominator 2^{512} : centres A_0, B_0, J_0 , a matrix X , $W_0 \approx \mathcal{W}(X)$, $Y \approx W_0^{-1}$, and congruence transforms for $C = B - mA$, W and T . The checker does not import the proposer and computes no eigenvalues. It verifies:

- all entries against the saved source balls, whose SHA-256 sums are pinned;
- $\epsilon_J = 2^{-250}$ (quadrature, the δ change and rounding), $\epsilon_W = 2^{-350}$ and $\epsilon_D = 2^{-380}$;
- the inverse residual, measured by the geometric mean of the row-sum and column-sum norms;
- $\eta + \epsilon_T \|Z\|_F^2 < 1$, which gives $\lambda_{\min}(T) \geq (1 - \eta - \epsilon_T \|Z\|_F^2) / \|Z\|_F^2$, where $\epsilon_T = 2b^2\epsilon_W + \epsilon_Y + \epsilon_J + \beta\epsilon_D$.

	even	odd
total error ϵ_T	5.53×10^{-76}	5.53×10^{-76}
certified lower bound for $\lambda_{\min}(T)$	$1.2092639998 \times 10^{-54}$	$1.5867129002 \times 10^{-50}$

TABLE 2. The certified finite condition (3.4) at $N = 832$, $\mu = 12$.

The checker passed at 1280 and at 1536 bits. It rejected seven altered inputs for the intended reasons: the sign of J reversed; an inflated declared error; B replaced by A^2 ; a file changed without updating its hash; an understated error of J ; a wrong contract; and a negative T . An independent reimplementaion of the final check at 1664 bits, which does not share code with the checker, reproduced both lower bounds to more than 90 digits; it also recomputed nine further source entries independently. The checker runs in about 32 seconds and 1.3 GB of memory.

Proof of Theorem M. The certificate gives (3.4) in both parities, hence $G^{-1} + J \succeq \alpha I + \beta D_M$ because $\mathcal{W}(X) \succeq G$. By (3.3), $M + V \succeq 0$ on H . By (3.2), $Q(f) \geq C\delta \|f\|^2 \geq 2^{-1518} \|f\|^2$. $\square$

6. PROOF OF COROLLARY 1.1

In [14] ($L = \frac{1}{2} \log 10$, $\Omega = 256$, $\gamma = \frac{7}{2}$, $m = \frac{1}{2}$, $N = 640$, $u = (81, 27)$), the out-of-band constant was $\delta_{\text{old}} = 2^{-49158}$, and the certified finite margin was above 2^{-144} in both parities. By Theorem N with $c = 128 \log 10 \approx 294.73$ we have $-\log_2 d(c) = 871.73 \dots$, so $\delta = 2^{-872} \leq d(c)$ is admissible.

Replacing δ_{old} by δ changes only U : with $t = \delta - \delta_{\text{old}}$, $h_j^{\text{new}} = h_j^{\text{old}} - tv_j$, and $\|U^{\text{new}} - U^{\text{old}}\| \leq 108(2t + t^2) < 2^{-863}$. The constant δ enters nothing else in [14]: not A , B , $\mathcal{W}$, nor the error budgets. Moreover $(I - P)h_j$ is unchanged, and $\|h_j^{\text{new}}\|^2 \leq e_j$, so the bounds e, n, h remain valid. The finite condition therefore keeps a margin above $2^{-144} - 2^{-863} > 2^{-145}$.

The tail inequality of [14, Proposition 3.4] holds with δ in place of δ_{old} : the only properties used were $I - B_{\text{band}} \succeq \delta I$, $e_j > \delta$ and $C_0\delta > \delta/16$. Hence $Q(f) \geq \delta/16 \|f\|^2 = 2^{-876} \|f\|^2$.

7. REMARKS

Larger windows. Theorem N supplies an out-of-band estimate for every $L > 0$, replacing the interpolation estimate stated in [14] under the convenient sufficient condition $L < 7/6$. The obstacle for larger windows is now the prime operator. As e^{2L} grows, s grows, hence Ω (through $C > 0$), N , and the precision. Rough estimates suggest $\Omega \approx 600$ – 900 for $e^{2L} = 15$. For $e^{2L} = 20$ the expected margin is about 10^{-96} , which requires error budgets below 2^{-340} . Both need a more economical implementation than ours.

Margins. The certified finite margins are of the same order as the numerical bottom of the spectrum. The final constants 2^{-1518} and 2^{-876} are much smaller because they come from the out-of-band bound, which is of size $e^{-2\Omega L}$.

Trust. The proof relies on the analytic lemmas above and in [14], on Arb/FLINT [1] for the ball enclosures, and on the checker program. It has not been checked by a proof assistant.

REPRODUCIBILITY

The source matrices, the certificate, the proposer, the checker and the mutation tests are in the folder `certificate-VII/` of release `v1.5.0` of <https://github.com/moriryota/prolate-weil-bounds> (about 350 MB), with SHA-256 sums and replay instructions, archived on Zenodo, doi:10.5281/zenodo.23092422 (all versions). The checker runs in about 32 seconds and 1.3 GB of memory.

USE OF GENERATIVE AI

This work was carried out with extensive use of generative AI systems, namely Anthropic Claude and OpenAI GPT. Under the direction of the author, these systems derived the analytic bounds, implemented and ran the computations, and reviewed one another's work adversarially. The author is responsible for the content.

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